

Homework 2

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1. We will need the disjoint unions

$$\begin{aligned}(A \cup B) &= (A \cap B^c) \cup (B \cap A^c) \cup (A \cap B), \\ A &= (A \cap B) \cup (A \cap B^c), \\ B &= (B \cap A) \cup (B \cap A^c),\end{aligned}$$

which imply

$$\begin{aligned}P(A \cup B) &= P(A \cap B^c) + P(B \cap A^c) + P(A \cap B), \\ P(A) &= P(A \cap B) + P(A \cap B^c), \\ P(B) &= P(B \cap A) + P(B \cap A^c).\end{aligned}$$

Also the independence assumption gives $P(A \cap B) = P(A)P(B)$. Thus we have

$$\begin{aligned}P(A^c \cap B^c) &= P((A \cup B)^c) = 1 - P(A \cup B) \\ &= 1 - [P(A \cap B^c) + P(B \cap A^c) + P(A \cap B)] \\ &= 1 - [(P(A) - P(A \cap B)) + (P(B) - P(A \cap B)) + P(A \cap B)] \\ &= 1 - P(A) - P(B) + P(A \cap B) \\ &= 1 - P(A) - P(B) + P(A)P(B) \\ &= (1 - P(A))(1 - P(B)) \\ &= P(A^c)P(B^c).\end{aligned}$$

So A^c and B^c are independent.

2. First write B as a disjoint union

$$B = (B \cap A) \cup (B \cap A^c),$$

which implies

$$P(B) = P(B \cap A) + P(B \cap A^c).$$

Thus we can write

$$P(A|B) = \frac{P(B \cap A)}{P(B)} = \frac{P(B) - P(B \cap A^c)}{P(B)} = 1 - P(A^c|B).$$

3. Let the sample space be all members, and set

$$\begin{aligned}A_1 &= \{\text{Members over the age of 65}\}, \\ A_2 &= \{\text{Members between the ages of 40 and 65}\}, \\ A_3 &= \{\text{Members under the age of 40}\}, \\ C &= \{\text{Members who have hospital expenses exceeding \$100,000}\}.\end{aligned}$$

We know $P(A_1) = 0.2$. Let $p = P(A_3)$ be the percentage of under-40 population. Since A_1 , A_2 and A_3 are disjoint and their union is the sample space, we can write $P(A_2) = 1 - P(A_1) - P(A_3) = 0.8 - p$. By the Law of Total Probability,

$$\begin{aligned} P(C) &= P(C|A_1)P(A_1) + P(C|A_2)P(A_2) + P(C|A_3)P(A_3) \\ &= (0.1)(0.2) + (0.04)(0.8 - p) + (0.03)(p) \\ &= 0.052 - 0.01p. \end{aligned}$$

For $P(C)$ to be less than or equal to 0.05, we must have $0.052 - 0.01p \leq 0.05$, or equivalently $0.2 \leq p$. So at least 20% of all members must be under 40 for HMO to avoid exceeding the trigger point.

4. Set

$$\begin{aligned} B_0 &= \{\text{Chosen coin has tails on both sides}\}, \\ B_1 &= \{\text{Chosen coin is fair}\}, \\ B_2 &= \{\text{Chosen coin has heads on both sides}\}, \\ A &= \{\text{Toss comes up heads}\}. \end{aligned}$$

The desired (conditional) probability is

$$\begin{aligned} P(B_1|A) &= \frac{P(A|B_1)P(B_1)}{P(A|B_0)P(B_0) + P(A|B_1)P(B_1) + P(A|B_2)P(B_2)} \\ &= \frac{(1/2)(1/3)}{(0)(1/3) + (1/2)(1/3) + (1)(1/3)} \\ &= 1/3. \end{aligned}$$

5. Since A_1 and A_2 are independent conditional on B , we have $P(A_1 \cap A_2|B) = P(A_1|B)P(A_2|B)$, or equivalently,

$$\frac{P(A_1 \cap A_2 \cap B)}{P(B)} = \frac{P(A_1 \cap B)}{P(B)} \frac{P(A_2 \cap B)}{P(B)}.$$

Thus we can write

$$\begin{aligned} P(A_2|A_1 \cap B) &= \frac{P(A_1 \cap A_2 \cap B)}{P(A_1 \cap B)} \\ &= \frac{P(A_1 \cap A_2 \cap B)}{P(B)} \frac{P(B)}{P(A_1 \cap B)} \\ &= \frac{P(A_1 \cap B)}{P(B)} \frac{P(A_2 \cap B)}{P(B)} \frac{P(B)}{P(A_1 \cap B)} \\ &= \frac{P(A_2 \cap B)}{P(B)} \\ &= P(A_2|B). \end{aligned}$$